Wulff Equilibrium Shapes of Cu and Pd Nanoparticles

Copper (Cu) and Palladium (Pd) are transition metals frequently used as catalytically active phases for a large diversity of reactions, including *e.g.* the NO_x reduction of exhaust gases, as carried out by Pd nanoparticles (NPs) supported on an Al_2O_3 porous and inert substrate,[1] or the water gas shift reaction, carried out by Cu NPs supported on ZnO.[2] In all these situations, Cu and Pd NPs are used in order to maximize their weight efficiency maximizing the exposed area.

At this point one may wonder what sorts of surfaces are exposed on such NPs, and to what extent, as their featuring and extent may imply different catalytic performances. One simple approach is to find the equilibrium shape of such NPs, obtained through the Wulff procedure of minimizing the overall NP surface energy (surface tension). The Wulff construction implies that, knowing the surface energy (γ) of a different set of surfaces, γ_n, the following equivalency has to be followed

$$\frac{\gamma_1}{h_1} = \frac{\gamma_2}{h_2} = \frac{\gamma_3}{h_3} = \cdots \quad (1),$$

where h_n are the shortest distances of the NP facets to the NP centre, see Figure 1.

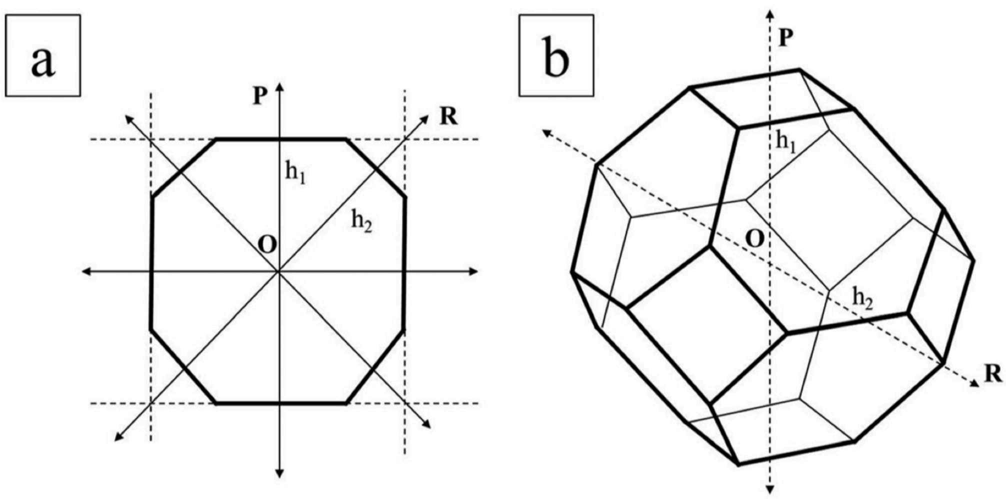

Figure 1. Example of a 2D (a) and 3D (b) Wulff shape.[3]

The necessary ingredients to depict a Wulff shape is a set of equivalent {*hkl*} Miller index surfaces, their symmetry, and their surface energies. In the case of Cu and

[1] Viñes *et al. J. Phys. Chem. C* 112, **2008**, 16539–16549.
[2] Zhang *et al. Nat. Commun.* 8, **2017**, 488.
[3] Viñes *et al. Chem. Soc. Rev.* 43, **2014**, 4922–4939.

Pd, being both face-centred cubic (*fcc*) metals, we will explore the most stable {001}, {011}, and {111} surfaces, see Figure 2.

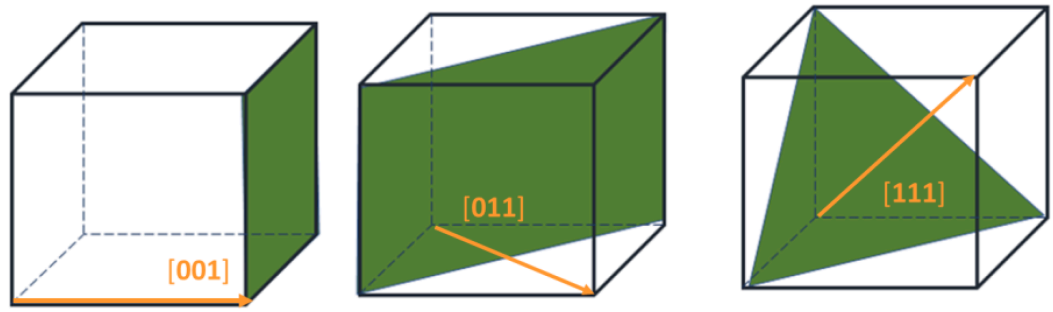

Figure 2. Examples of (001), (011), and (111) surfaces for a cubic crystal, such as *fcc* transition metals.

For each surface and metal, slab models have been constructed containing six atomic layers, and a minimum and necessary vacuum. These layers have been built from a previous bulk optimization, although have been later on fully relaxed. The calculations have been carried out using the Vienna *Ab Initio* Simulation Package (*VASP*) code.[4] With the supplied information, one can extract the fixed surface energy, γ^{fix}, defined as;

$$\gamma^{fix} = \frac{(E_{slab}^{fix} - nE_{bulk})}{2A} \qquad (2),$$

where E_{slab}^{fix} is the energy of the slab without optimization, E_{bulk} the energy of the optimized bulk, A the area of a created surface. As well, one can acquire the surface energy of the relaxed system, γ^{fix}, as;

$$\gamma^{relax} = \frac{(E_{slab}^{relax} - nE_{bulk})}{2A} \qquad (3),$$

where E_{slab}^{relax} is the energy of the slab after relaxation.

Such information can be used to build up the Wulff shapes, with a series of bulk and surface optimizations, which are already supplied. To create the Wulff shape, we will use the Visualization for Electronic and Structural Analysis (VESTA) program,[5] which you will have to install in your machine.

The basics imply loading the bulk optimized structure, and later shaping the NP, by using the Edit > Edit Data > Crystal Shape module. Here we can add New planes, and put them manually or exploiting the crystal Symmetry. Notice that surfaces have to be added following the Wulff rule, as in Eq. 1. After adding the surfaces, you can

[4] https://www.vasp.at
[5] http://jp-minerals.org/vesta/en/

Customize Colour, even to make such surfaces semi-transparent. Finally, one can delete the bulk template model, by Not Showing it. Once can Save As the image in VESTA format, even Save Images. Note that by clicking a given surface, one can get a value of its area in the bottom dialogue part of VESTA.

Tasks

- Explain the type of simulations and give model and computational details
- List the fixed and relaxed surface energies of Pd and Cu (001), (011), and (111) surfaces, alongside the relaxation energies. Which surface is the most stable for each material? Is relaxation a determining factor in the stability of the most stable surface?
- Acquire the Wulff shapes for Pd and Cu, using either fixed or relaxed surface energies. Show the percentages of exposed surfaces. Is relaxation playing a determining role in the Wulff shape? Which surfaces does one needs to considering when willing to study the catalytic activity on such NPs?
- BONUS TRACK #1: Determine the interlayer relaxation distance of all the studied surfaces. Was bulk Cu and Pd bonding attractive or repulsive?
- BONUS TRACK #2: Determine the Wulff shapes when using the *ab initio* atomistic thermodynamics to acquire the surface free energies instead of total surface energies.